

South African Bank Note Recognition

COMP702 — Image Processing and Computer Vision

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1 Introduction

Automatically identifying currency has many practical uses, including ATMs, vending machines, and assistive tools for visually impaired people. The difficulty lies in the fact that images of notes can vary substantially depending on how they were photographed: lighting conditions, camera angle, scale, and which side of the note is facing up all affect their appearance. Traditional approaches rely on hand-crafted image features [5], while more recent work tends to use convolutional neural networks. Both approaches can be effective, but a hand-crafted pipeline is often more practical when a large labelled dataset is not available [4].

For this project, we developed a pipeline to classify South African banknotes into five denominations: R10, R20, R50, R100, and R200, covering both the old Mandela series and the new Big Five series. A key requirement of the project was that the system must be invariant to which side of the note is shown (front or back), to the scale at which it appears in the image, and to its orientation. We achieved this through a combination of fixed-size resizing at load time (scale invariance) and data augmentation with flips and 90/180/270 degree rotations during training (side and rotation invariance). This means the classifier learns to recognise notes regardless of orientation or which face is shown, rather than relying on the note being upright. Scale invariance is handled separately at the input stage: all images are resized to 224×224 pixels on load, so the classifier sees a consistent scale regardless of how close or far the note was when photographed.

The pipeline follows four stages: preprocessing and enhancement, segmentation, feature extraction, and classification. We extracted HOG, LBP, and colour histogram features and compared three classifiers to determine which performed best. Our dataset started at 154 images and was expanded to 1386 through augmentation.

The code for this project is available at (see the included README for setup and execution instructions): <https://github.com/KgFix/South-African-Bank-Notes-Recognition>

2 Image Preprocessing and Enhancement

Before running any analysis, all images were resized to 224×224 pixels on load to ensure consistent dimensions across the dataset. Three preprocessing steps were then applied in sequence.

Grayscale Conversion. Each image was converted from three RGB channels down to a single grayscale channel. Colour information was still used later in the pipeline through a dedicated colour histogram feature, but for the structural descriptors (HOG and LBP) grayscale was sufficient and kept the computation simpler.

CLAHE (Contrast-Limited Adaptive Histogram Equalisation) was applied after the grayscale conversion. Some images in the dataset were captured under uneven lighting, which caused parts of the note to appear washed out or too dark. CLAHE addresses this by improving contrast locally across small tiles of the image rather than globally, which avoids over-amplifying noise in regions that are already well-exposed.

Gaussian Blur was the final step. A mild Gaussian blur was applied to reduce sensor noise before the edge-based and texture-based analysis in later stages.

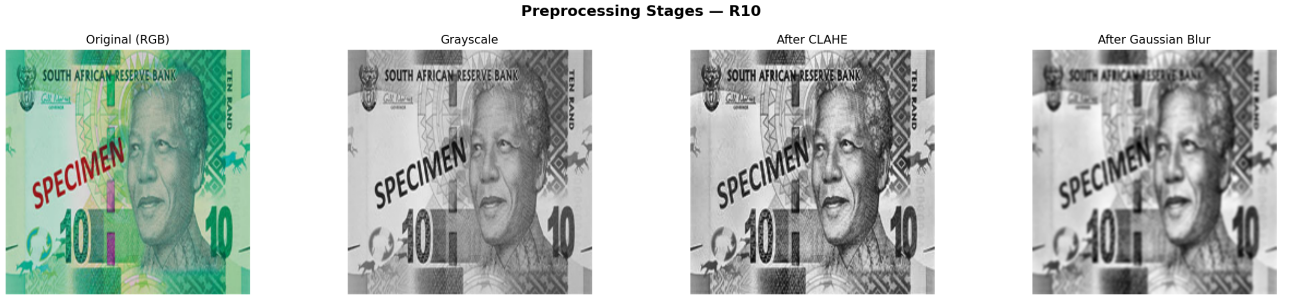


Figure 1: The four preprocessing stages applied to one example image: original (RGB), grayscale, after CLAHE, and after Gaussian blur.



Figure 2: One sample image per denomination from the original dataset, illustrating the colour-coding used to distinguish each note: R10 (green), R20 (brown/tan), R50 (red), R100 (blue), and R200 (orange-gold). This colour distinction is the basis for the colour histogram feature used in classification.

Data Augmentation. Our original dataset contained 154 images across the five denominations (R10: 32, R20: 32, R50: 32, R100: 30, R200: 28) [6] [7], which is quite small for training a classifier reliably. To increase the training data and to build rotation and side invariance directly into the model, we generated eight augmented variants per image: a horizontal flip, a vertical flip, three rotations (90° , 180° , 270°), two brightness adjustments using alpha/beta scaling, and one version with added Gaussian noise. This brought the total dataset to 1 386 images. It should be noted that augmentation was applied before splitting into train and test sets, which means augmented copies of training images appear in the test set. This inflates the reported test accuracy, so the cross-validation results in Section 5 are a more honest measure of real performance.

3 Image Segmentation

The goal of segmentation was to isolate the bank note from its background so that feature extraction would focus on the note itself rather than the surface it was photographed on. We tried two methods.

Otsu’s Thresholding [3] automatically determines the best threshold value for separating the foreground (the note) from the background by analysing the image histogram. The main advantage is that it requires no manual parameter tuning and adapts to each image individually.

Canny Edge Detection with Contour Extraction takes a different approach by tracing the boundary of the note, using morphological closing to fill gaps in the outline, and then filling the largest closed contour to produce a binary mask. This method can perform well when there is a strong contrast between the note and the background, but it is more sensitive to parameter choices and tends to struggle when that contrast is low.

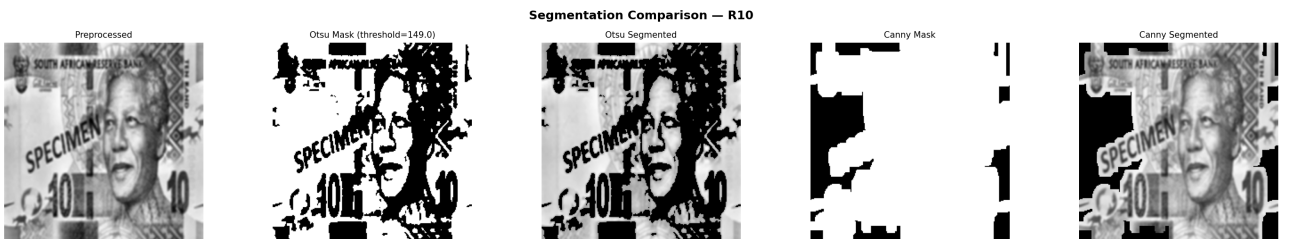


Figure 3: Segmentation results for the same image using both methods: original, Otsu’s mask, and the Canny-based mask.

We chose Otsu’s thresholding as our primary segmentation method. Across the dataset it produced cleaner and more consistent masks, and was much less sensitive to variation in background colour and lighting. The CLAHE step from preprocessing feeds directly into this, since improved local contrast makes the histogram bimodal distribution that Otsu relies on clearer and easier to threshold.

4 Feature Extraction

For each segmented image, we extracted three types of features and concatenated them into a single feature vector of 6 206 dimensions. The breakdown is as follows: HOG contributes the majority at approximately 6 084 values, LBP contributes a 26-bin normalised histogram, and the colour histogram contributes 96 values (32 bins per RGB channel).

HOG (Histogram of Oriented Gradients) [1] captures the shape and edge structure of a note by measuring how image gradients are oriented across small local regions. We used 9 gradient orientations, 16×16 pixel cells, 2×2 cell blocks, and L2-Hys block normalisation.

LBP (Local Binary Patterns) [2] captures local texture by comparing each pixel to its circular neighbourhood and encoding the comparison as a binary pattern. We used a larger-radius uniform LBP configuration with $P = 24$ neighbours at radius $R = 8$ pixels (26-bin histogram) to capture the coarser printed patterns on the notes rather than fine-grain camera noise. The histogram was normalised to sum to 1 so that results are not affected by image size.

Colour Histogram. This turned out to be one of the most discriminative features, as South African notes are colour-coded by denomination: R10 is green, R20 is brown/tan, R50 is red, R100 is blue, and R200 is orange-gold. We extracted a 32-bin histogram for each of the three RGB channels and concatenated them. Using HSV colour space would separate hue from brightness more cleanly and could help in variable lighting conditions, which is something we would explore if extending this work.

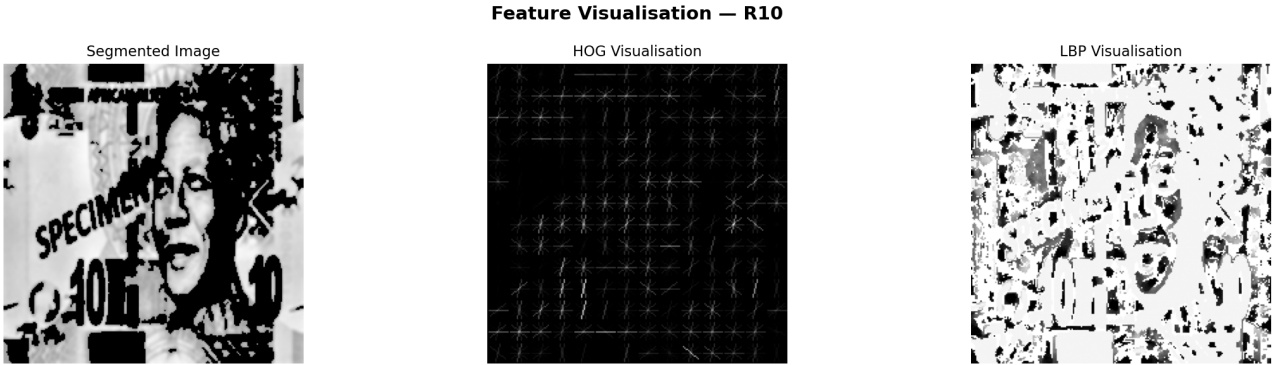


Figure 4: HOG gradient map and LBP texture map for one example image.

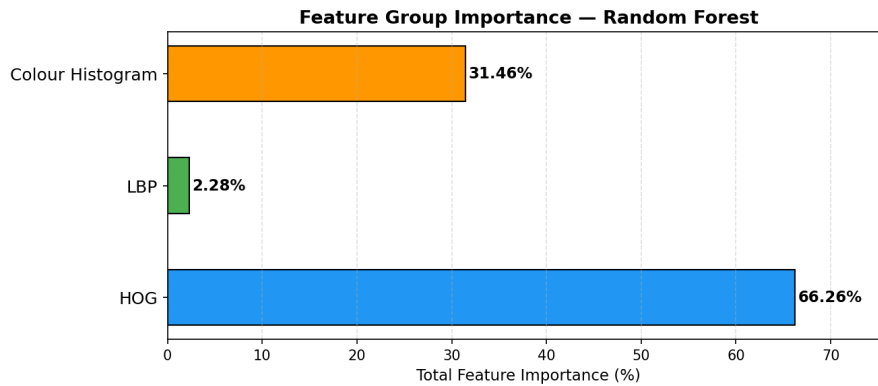


Figure 5: Random Forest feature group importances. The colour histogram contributes the largest share of predictive power, consistent with each denomination having a distinct dominant colour. HOG captures the structural layout of each note and contributes the second-largest share.

After concatenating all three feature sets into the 6 206-dimensional vector, we standardised everything to zero mean and unit variance using a `StandardScaler` [4]. The scaler was fitted only on the training

data to prevent any test-set information from leaking into the feature normalisation.

5 Note Classification

We trained three classifiers on the extracted feature vectors and evaluated them on an 80/20 stratified train-test split. We also ran 5-fold cross-validation on the full augmented dataset. As noted in Section 2, the test set accuracy is inflated due to the augmentation-before-split approach, so the cross-validation mean is the more reliable indicator of how the system would perform on genuinely unseen images.

SVM with an RBF kernel ($C = 10$, `gamma='scale'`) was one of our two main classifiers. SVMs tend to work well in high-dimensional feature spaces and produce probability outputs that we used for confidence scoring in the prediction tool.

Random Forest with 200 estimator trees was our second main classifier. By aggregating predictions across many decision trees it is robust to noisy or irrelevant features, and it does not require feature scaling the way distance-based methods do.

KNN with $k = 5$ and distance weighting was included as a baseline. We expected it to underperform relative to the other two classifiers because nearest-neighbour methods are known to struggle in very high-dimensional spaces. With 6 206 features, Euclidean distances between points become less informative, which is a well-known consequence of the curse of dimensionality.

Results

Table 1: Results on the hold-out test set and 5-fold cross-validation. The CV mean is the more reliable indicator of real-world performance given the augmentation leakage in the test set.

Classifier	Test Acc.	Precision	Recall	CV Mean $\pm$ Std
Random Forest	98.92%	0.99	0.99	87.38% $\pm$ 8.99%
Support Vector Machine	98.20%	0.98	0.98	88.97% $\pm$ 12.10%
K-Nearest Neighbours	88.13%	0.88	0.88	79.02% $\pm$ 14.22%

On the test set, Random Forest achieved the highest accuracy at 98.92% with SVM (Support Vector Machine) close behind at 98.20%. The cross-validation results tell a slightly different story: SVM generalises better with a CV mean of 88.97% compared to 87.38% for Random Forest. The wide standard deviation for SVM (12.10%) reflects how strongly the augmentation leakage affects fold scores, with individual folds ranging from 69.3% to 100.0%. KNN performed noticeably worse as expected, with a CV mean of 79.02%, confirming that high-dimensional feature spaces are not suited to nearest-neighbour approaches.

Looking at the confusion matrices, R20 and R50 were the most commonly confused pair across all three classifiers. This is most likely because their colours can appear similar under certain lighting conditions, which would be addressed by switching to HSV histograms.

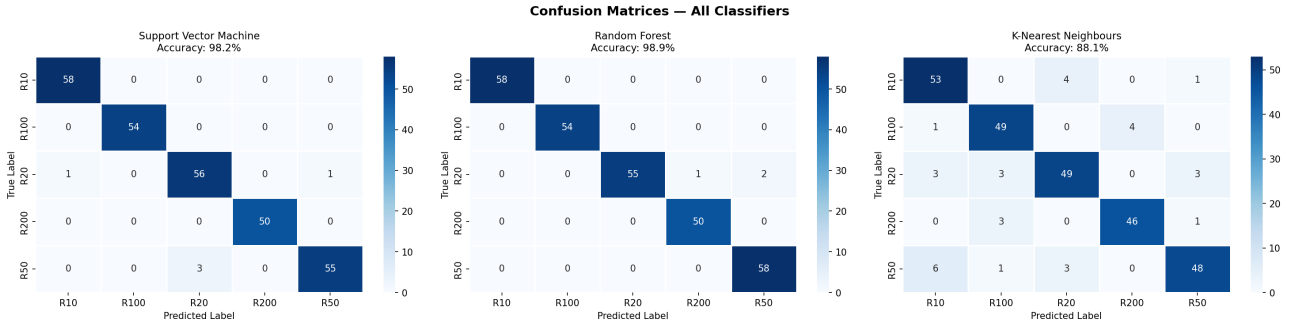


Figure 6: Confusion matrices for all three classifiers on the hold-out test set.

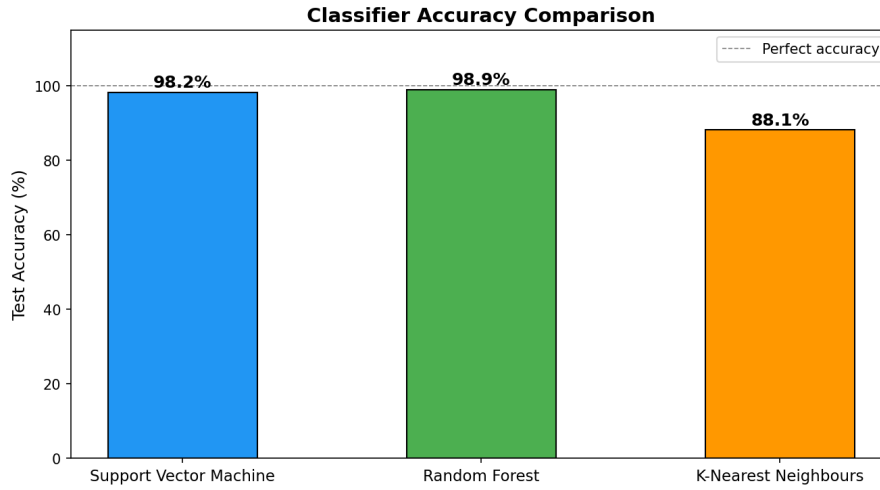


Figure 7: Test accuracy and 5-fold CV mean comparison across the three classifiers.

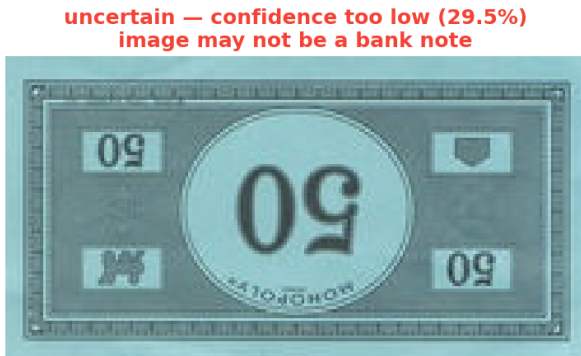
We added a confidence threshold to the live prediction tool: if the top class probability is below 50%, the system flags the result as uncertain rather than returning a potentially wrong classification. This makes the tool more reliable when fed images that are not bank notes or where the note is heavily obscured.



(a) R100 correctly classified at 93.0% confidence (green).



(b) R200 correctly classified upside down at 65.5% confidence (orange).



(c) Monopoly money rejected — confidence below 50% threshold (red).



(d) Fake note rejected — confidence below 50% threshold (red).

Figure 8: Live prediction tool outputs across four test cases. The R100 is classified at 93.0% confidence and the R200 is correctly classified despite being upside down at 65.5% confidence, demonstrating the rotation invariance built in through augmentation. Both the Monopoly money and fake note fall below the 50% threshold and are flagged as uncertain, demonstrating robustness to non-genuine inputs.

6 Conclusion

The pipeline performed well overall. CLAHE handled lighting variation better than we expected, Otsu’s thresholding produced clean segmentation without requiring parameter tuning, and the combination of HOG, LBP, and colour histograms gave the classifiers enough discriminative information to achieve close to 99% on the test set and around 87–89% on cross-validation.

There are several things we would change given more time. The most significant issue is the augmentation issue mentioned earlier. Because augmentation was done before the train-test split, augmented copies of training images ended up in the test set, inflating those accuracy figures. The cross-validation results (RF: $87.38\% \pm 8.99\%$, SVM: $88.97\% \pm 12.10\%$) are a much more realistic picture of actual performance. The correct approach would be to apply augmentation only within each training fold, which requires a custom pipeline wrapper in scikit-learn.

A second limitation is that the dataset consists mostly of flat, well-lit images. When we tested the system on photos of someone actually holding a note, accuracy dropped noticeably. In real holding conditions the note is smaller in the frame, at an angle, the background is cluttered, and parts of the design are covered by hands, none of which the model was trained to handle.

The R20/R50 confusion is also worth addressing. Switching from RGB to HSV colour histograms would separate the hue channel from brightness and likely reduce this specific error.

In the longer term, replacing the hand-crafted feature pipeline with a CNN trained on a more diverse and realistic dataset would probably be the most effective solution to all three of these limitations.

References

- [1] Dalal, N., & Triggs, B. (2005). Histograms of oriented gradients for human detection. *Proc. IEEE CVPR*, 1, 886–893.
- [2] Ojala, T., Pietikainen, M., & Maenpaa, T. (2002). Multiresolution gray-scale and rotation invariant texture classification with local binary patterns. *IEEE Trans. Pattern Anal. Mach. Intell.*, 24(7), 971–987.
- [3] Otsu, N. (1979). A threshold selection method from gray-level histograms. *IEEE Trans. Syst. Man Cybern.*, 9(1), 62–66.
- [4] Pedregosa, F. et al. (2011). Scikit-learn: Machine learning in Python. *J. Mach. Learn. Res.*, 12, 2825–2830.
- [5] Gonzalez, R.C., & Woods, R.E. (2002). *Digital Image Processing* (2nd ed.). Prentice Hall.
- [6] South African Reserve Bank. (2025). *Banknotes and Coin: Banknotes*. Retrieved from <https://www.resbank.co.za/en/home/what-we-do/banknotes-and-coin/Banknotes>
- [7] BanknoteDB. (n.d.). *South Africa — South African Reserve Bank banknote catalogue*. Retrieved from <https://banknotedb.com/?country=248&bank=500>